

STATEMENT OF WORK

SECTION C – DESCRIPTION/SPECIFICATION/WORK STATEMENT

C.1 BACKGROUND

Naval Surface Warfare Center Dahlgren Division Dam Neck Activity (NSWCDD DNA) is assigned the duties of systems and software engineering, development, testing, hardware engineering, site support, Life-Cycle support inclusive of Integrated Logistics Support (ILS) and Configuration Management (CM) activities, fielding and maintenance support, In-Service Engineering Agent (ISEA), Cybersecurity (Cyber), C6ISR (Command, Control, Communications, Computers, Cyber-Defense and Combat Systems and Intelligence, Surveillance and Reconnaissance), Safety, Test and Evaluation (T&E), and Readiness support for assigned shore based training systems.

These system assignments include the hardware, software, subsystems, and elements that comprise the specific system. These activities are conducted in support of the United States (U.S.) Navy, Multi-National and Multi-Agency Naval and Department of Defense (DoD) sponsors.

The Integrated Training Systems (ITS) division is the primary support element of program office shore-based training objectives within NSWCDD DNA. The ITS division serves as a leader in shore-based training research, engineering, development, modernization, sustainment, integration and interoperability of Navy, Joint and Coalition surface combat shore-based training systems.

The capabilities it provides encompasses modeling and simulation (M&S), evaluation and testing as well as other diverse information technology approaches to include hardware and software elements specific to shore-based training and readiness systems.

C.2 SCOPE

The scope of the Task Order is to procure Contractor support services to achieve the acceptable levels of requirements analysis; software engineering; hardware engineering; systems engineering and technical support; software development and methodologies including development, security, and operations (DevSecOps) within an agile framework; software, hardware and systems integration; installation; testing and verification support; Cybersecurity and cyber engineering; lab support, outfit and evaluation; ILS and CM support; Life-Cycle management; provisioning support; material disposition and inventory management. Such support shall include support of the U.S. Navy, Multi-National and Multi-Agency software development efforts in the ITS Division for the modernization and sustainment of shore-based training systems.

C.2.1 Applicable Documents

The following documents are applicable to this Statement of Work (SOW). The most current versions of these documents are accessible through the internet or will be provided to the Contractor upon request.

Other Department of Defense Specifications/Standards/Other Applicable Documents may become applicable throughout the life of the Task Order and may be added to the Task Order via modification.

The Contractor shall ensure compliance with the most current versions of these documents throughout the life of the Task Order:

Number	Document Name
MIL-STD-31000B	Standard Practice Technical Data Packages
MIL_STD-38784A CHG 2	DoD Standard Practice for Manuals, Technical: General Style and Format Requirements
MIL-STD-2073-1E CHG 4	Standard Practice for Military Packaging
MIL-STD-129R CH 2	Military Marking for Shipment and Storage
MIL-DTL-24784D	Detail Specification: Manuals, Technical: General Acquisition and Development Requirements, General Specification
MIL-STD-46855A	DoD Standard Practice, Human Engineering Requirements for Military Systems, Equipment and Facilities
MIL-STD 2525D	DoD Interface Standard: Joint Military Symbolology
DoD 8570.01-M CH 4	Information Assurance Workforce Improvement Program
DoDI 8510.01 CH 3	Risk Management Framework (RMF) for DoD Information Technology (IT)
DoDI 8530.01	Cybersecurity Activities Support to DoD Information Network Operations
NSWCDD Instruction 5239.4C	Cyber Workforce Qualification Program
N/A	Naval Open Architecture Contract Guidebook for Program Managers
DN-STAVE-CS-SEP-19-0006	Systems Engineering Plan (SEP)
DN-STAVE-CS-QMP-20-0001	Quality Management Plan (QMP)
STAVE-CS LCSP-VMTT-PTT	Lifecycle Management Plan (LMP)
STAVE-CS-VMTT SDP Ver 1.0	Software Development Plan (SDP)
DN-STAVE-CS-CSP Rev. 2	Cyber Security Plan (CSP)
STAVE-CS CMP-VMTT-PTT	Configuration Management Plan (CMP)
N/A	Software Test Plan (STP)
N/A	Human Systems Integration Plan (HSIP)
N/A	STAVE-CS Interoperability Standard (SIS)
22 CFR 120-130	International Traffic in Arms Regulation (ITAR)
NAVSEAINST 5000.8	Naval SYSCOM Risk Management Policy

PEO IWS and NAVSEA	Technical Review Manual (TRM), Version 2.0
PEOIWSINST 5000.8	Memo PEO IWS Risk Management Policy Memo with (Enclosure 1: Instructions for Risk Reporting)
SECNAV 5239.3C	Department of the Navy (DoN) Cybersecurity Policy
SECNAV 5239.22	DoN Cybersecurity Safety Program (CYBERSAFE)
SECNAV M-5239.2	DoN Cyberspace Information Technology and Cybersecurity Workforce Management and Qualification Manual
SECNAV M-5239.3	Department of the Navy Cyber Security Manual
SECNAV M-5510.1	DoN Foreign Disclosure Manual
COMUSFLTFOR-COMINST 4790.3 Rev. D	Joint Fleet Maintenance Manual
ASME Y14.100-2017	Engineering Drawing Practices – Engineering Drawing and Related Documentation Practices
IEEE Std 1012-2016	System, Software, and Hardware Verification and Validation
IEEE Std 1278.1-2012	Distributed Interactive Simulation – Application Protocols
IEEE Std 1516.1-2010	Standard for Modeling and Simulation (M&S) High Level Architecture (HLA) – Federate Interface Specification
ISO/IEC/IEEE 1028-2008	Software Reviews and Audits
ISO/IEC/IEEE 12207-2017	Systems and Software Engineering – Software Life-Cycle Processes
ISO/IEC/IEEE 24774-2021	Systems and Software Engineering – Life-Cycle Management – Guidelines for Process Description
ISO/IEC/IEEE 26512-2017	Systems and Software Engineering – Requirements for Acquirers and Suppliers of Information for Users
NIST SP 800-12 Rev 1	An Introduction to Information Security
NIST SP 800-30 Rev 1	Guide for Conducting Risk Assessments
NIST SP 800-35	Guide to Information Technology Security Services
NIST SP 800-37 Rev 2	Risk Management Framework for Information Systems and Organizations: A system Life-Cycle Approach for Security and Privacy
NIST SP 800-39	Managing Information Security Risk: Organization, Mission, and Information System View
NIST SP 800-53 Rev 5	Security and Privacy Controls for Information Systems and Organizations
NIST SP 800-115	Technical Guide to Information Security Testing and Assessment
NISP SP 800-160 Vol. 1	Engineering Trustworthy Secure Systems
NISP SP 800-160 Vol.2	Developing Cyber Resilient Systems: A Systems Security Engineering Approach

NISP SP 800-161	Supply Chain Risk management Practices for Federal Information Systems and Organizations
NISPOM – 32 Code of Federal Regulation Part 117	National Industrial Security Program Operating Manual (NISPOM)
NSWCDDINST 5510.4B	NSWCDD Command Security Manual
SEAD-3	Security Executive Agent Directive 3 – Reporting Requirements for Personnel with Access to Classified Information or Who Hold a Sensitive Position
SEAD-3 Implementation Memo	Implementation of Security Executive Agent Directive 3, Reporting Requirements for Personnel with Access to Classified Information or Who Hold a Sensitive Position
DODI 5200.48	Controlled Unclassified Information
SECNAVINST 5210.8F	Department of Navy Records Management Program
NSWCDDINST 5100.1F	Occupational Safety and Health Program
NAVADMIN 216/15	Cyber Hygiene Authorization to use Personal Wearable Fitness Devices
NAVSEA S0420-AA-RAD-010 – REV 2A	Radiological Affairs Support Program (RASP) Manual
DODI 8510.01	Risk Management Framework for DOD Systems
CJCSI 6211.02D	Defense Information Systems Network (DISN) Responsibilities
DODM 1000.13	Department of Defense Identification (ID) Cards
NAVSEAINST 4200.19 (series)	Planning, Use and Control of Services Contracting
COMUSFLTFORCOM/COMPACFLT 6320.3B	Medical Screening for U.S. Government Civilian Employees, Contractor Employees, Guests and Visitors Prior to Embarking Fleet Units

C.3 REQUIREMENTS

- a. The Contractor shall provide all contract administration, contract management, personnel, tools, equipment, material, vehicles, transportation, consumable supplies, operation, and maintenance to accomplish the requirements of this SOW.
- b. The Contractor shall provide recommended solutions for technical Information Technology (IT) issues and perform to the standards, and the degree of ability, knowledge, skills, and timeliness required in the SOW. The Contractor shall have the capability of developing new technologies and have the technical expertise to transition into new operating environments.
- c. The Contractor shall accomplish the work and provide the deliverables specifically described in the SOW. Contract objectives are based on the SOW, which is organized by the following disciplines:
 - Software Engineering (C.3.1)
 - Software Development and Testing (C.3.2)
 - Database and Computer Systems Administration (C.3.3)

- Systems and Technical Engineering Services (C.3.4)
- Cybersecurity (C.3.5)
- Test and Evaluation (C.3.6)
- Life-Cycle Maintenance (C.3.7)
- Configuration Management (C.3.8)
- Lab Support (C.3.9)
- Shore Based Training System Hardware Engineering (C.3.10)
- C6ISR: Command, Control, Communications, Computers, Cyber-Defense and Combat Systems and Intelligence, Surveillance, and Reconnaissance (C.3.11)
- Material Analysis (C.3.12)
- Training Site Installation (C.3.13)
- Training Site Accreditation (C.3.14)
- Training Site Operations (C.3.15)

C.3.1 Software Engineering

The Contractor shall provide software engineering support services to the Government in support of shore-based training objectives.

- a. The Contractor shall review and analyze current and future requirements to identify potential changes, modifications and impacts to existing software product lines, training systems and associated systems/subsystems and hardware infrastructures.
- b. The Contractor shall identify, assess, and report cost, schedule and technical risks associated with all stages of the Software Lifecycle.
- c. The Contractor shall conduct research into advanced analytical tools, processes, and technologies for future software development and integration.
- d. The Contractor shall develop software quality assurance plans and procedures to ensure that products and services delivered are IAW approved standards and plans.
- e. The Contractor shall develop white papers, trade studies and decision option papers while leveraging industry standards to provide technical recommendations, feedback, and coding standards that lead to development of software programs that are scalable, portable, and maintainable.
- f. The Contractor shall provide software code, design, and test cost estimates and schedules for Initial Requirement Proposals (IRPs) and Engineering Change Proposals (ECPs).
- g. The Contractor shall review and provide analysis of Software Requirement Specifications (SRS), associated requirements documentation and system level designs to ensure requirements are traceable and align with end user/design specifications.
- h. The Contractor shall review and provide analysis of Interface Requirement Specifications (IRS), Interface Design Documentation (IDD), Weapon Specifications (WS) and other

interface control documentation to ensure the compliance of the software with approved specifications.

- i. The Contractor shall coordinate with stakeholders during software engineering and development efforts, including with international military designated contacts and other Agency partners.
- j. The Contractor shall develop software preliminary and detailed designs, including the development of Software Design Documents (SDDs) and associated design documentation.
- k. The Contractor shall develop pseudo code and Unified Modeling Language (UML) diagrams to support software design to include use case, class, sequence, and associated diagrams using Enterprise Architect (EA) or equivalent.
- l. The Contractor shall develop and maintain an Open Architecture (OA) design that incorporates re-configurability, portability, maintainability, technology insertion, vendor independence, reusability, scalability, interoperability, upgradeability, and long-term supportability as defined in the Navy Open Architecture (NOA) Contract Guidebook.

APPLICABLE CDRL(s):

C.3.2 Software Development and Testing

The Contractor shall provide software development and testing support services to the Government in support of shore-based training objectives.

- a. The Contractor shall develop computer programs that meet end user and system requirements while maximizing reusability of software. Computer programs will be written in one of the following computer programming languages: C, C++, C#, Java, Ada, JavaScript, PostgreSQL, and JQuery.
- b. The Contractor shall develop software that handles input and output in a variety of data format. These formats are flat files, Extensible Markup Language (XML), Experience API and JavaScript Object Notation (JSON).
- c. The Contractor shall develop software using tools in Integrated Development Environments (IDE).
- d. The Contractor shall develop interfaces with shore site components using protocols: Distributed Interactive Protocol (DIS), High Level Architecture (HLA) and Application Programming Interfaces (APIs).
- e. The Contractor shall configure and maintain Operating Environments (OE) in VxWorks, Windows Operating Systems (OS), and Linux Operating Systems (OS).

- f. The Contractor shall design and implement web services using Simple Object Access Protocol (SOAP) and the Representational State Transfer (REST) architectural approach.
- g. The Contractor shall design and develop software using gaming frameworks to support virtual reality (VR) training capabilities. It is expected that the preponderance of development will take place using Unity.
- h. The Contractor shall design, develop, install, configure and maintain software and web services while conforming to Cybersecurity standards and practices including Security.
- i. The Contractor shall prepare Security Technical Implementation Guides (STIGs), Information Assurance Vulnerability Alerts (IAVAs), and Assured Compliance Assessment Solution (ACAS) reports.
- j. The Contractor shall provide associated documentation for all source code.
- k. The Contractor shall perform code reviews to prior to delivering software artifacts.
- l. The Contractor shall develop milestone technical review, design, and test readiness presentations in accordance with the PEO IWS and NAVSEA Technical Review Manual (TRM).
- m. The Contractor shall automate processes whenever possible to improve efficiency of software development efforts and automate build processes.
- n. The Contractor shall support system build production including compilation of software, creation of executables, and development of installation solutions using automation utilities.
- o. The Contractor shall document and maintain software compilation processes and procedures.
- p. The Contractor shall implement DevSecOps practices throughout the software development lifecycle.

APPLICABLE CDRL(s):

C.3.3 Database and Computer Systems Administration

The Contractor shall provide and maintain database architecture, development, administration, and maintenance support services to the Government in support of shore-based training objectives.

- a. The Contractor shall develop new and maintain existing database schemas and databases using relational database management systems or data storage, retrieval, and report applications.
- b. The Contractor shall develop and maintain the Database Design Document (DBDD).

- c. The Contractor shall document database backup and recovery procedures.
- d. The Contractor shall design and develop callable statements and database scripts using Structure Query Language (SQL).
- e. The Contractor shall support research of the data sources and coordinate the manual and automated retrieval of data from outside agencies. The Contractor shall ensure data is not degraded during transmission from source to the target database.
- f. The Contractor shall transform data into the required format for ingestion into the database and perform data imports using custom scripts or database administrative tools.
- g. The Contractor shall provide database administrative services, user account setup, and optimization.
- h. The Contractor shall ensure the database is following all applicable DISA STIGs security standards.

APPLICABLE CDRL(s):

C.3.4 Systems and Technical Engineering Services

The Contractor shall provide systems and technical engineering support services through the implementation of System Engineering processes across all mission areas of Shore Based Training Systems.

- a. The Contractor shall perform engineering of systems' requirements, architectures, internal and external system interoperability communications, cybersecurity, software, hardware, integration, safety, deployment, and support across multiple systems' Lifecycles.
- b. The Contractor shall research, analyze, define, develop, prototype and review capabilities and requirements at the systems/subsystems level for integrated software and hardware products.
- c. The Contractor shall support the execution of engineering activities, develop technical approaches and plans that support the Stakeholder's desired capabilities. The Contractor shall assist with the determination of scope of work tasks to complete the technical effort and estimating time to accomplish the work tasks and establishing feasible schedules based on resource availability.
- d. The Contractor shall assist with the selection of performance measures and criteria against which technical progress can be assessed in terms of cost, schedule, and risk.
- e. The Contractor shall develop and implement approaches for the capture, analysis and trace of artifacts that provide the requirements of the system's performance, function and allocated baseline and maintain traceability from engineering products back to user-

defined capabilities as documented through the Joint Capabilities Integration and Development System (JCIDS), or equivalent.

- f. The Contractor shall support the operational equipment/user/operational test community to ensure the concept of operations (CONOPS) to verify and validate system performance and operational capability.
- g. The Contractor shall develop the design reference missions, use cases, and Unified Model Language (UML) model diagrams.
- h. The Contractor shall provide interface definition and compliance among the products that compose the system and with other systems with which training system products must operate.
- i. The Contractor shall participate in technical reviews to identify and document interface requirements and attributes between connected systems.
- j. The Contractor shall support the Government in assessment of technical risk, through risk identification and risk analysis. The Contractor shall provide risk mitigation options, and work with the Technical Manager to resource associated mitigation plans.
- k. The Contractor shall provide risk status and inputs to Program Reviews, Risk Management Boards, and System Engineering Technical Reviews.
- l. The Contractor shall support the development and application of policies, procedures, and information technologies to plan, acquire, access, manage, protect, and use technical data required by various training systems.
- m. The Contractor shall support formal and informal Systems Engineering Technical Reviews (SETRs).
- n. The Contractor shall host or attend Technical Interchange Meetings (TIMs), program Integrated Product Team (IPT) meetings, and working groups (WGs) which support training systems interfaces and SETR milestone reviews.
- o. The Contractor shall verify that products and artifacts conform to requirements. The Contractor shall identify, record, and remove defects and ensure any product line factors/issues are considered.
- p. The Contractor shall provide a basis for subsequent product approval at formal SETRs (e.g., System Requirements Review (SSR), Software Specification Review (SSR), System Functional Review (SFR), Preliminary Design Review (PDR), Critical Design Review CDR, and Test Readiness Review (TRR)).
- q. The Contractor shall support technical assessment activities that measure technical progress and effectiveness of plans and requirements. The Contractor shall review and assess baselines (functional, allocated, performance, product), CONOPS, capability descriptions, requirements, developers' preliminary and detailed designs, adequacy of

design, cost effectiveness, and provide recommendations to achieve cost avoidance where feasible; and evaluate developers' Plans of Action and Milestones (POA&Ms).

- r. The Contractor shall plan the execution of the integration of software and hardware components into the system, including analysis of desired capabilities, objectives, performance specification, and system design/architecture; development, maintenance and documentation of System Functional specifications and System Verification.
- s. The Contractor shall support shore site readiness in the areas of safety, maintenance engineering, computer program support, and Shore Site training policy as dictated by fleet operational training requirements and shore site requirements.
- t. The Contractor shall support shore site improvements in the following areas: hardware design and integration, human system integration, performance and maintenance data analysis, computer program improvements, new capabilities, and Pre-planned Product Improvement (P3I), maintenance of technical documentation specifications and standards, development of test equipment, tools, jigs, and fixtures to support improvements, and obsolescence support.
- u. The Contractor shall support shore site installation to ensure that planning and scheduling, site checks, drawing development, configuration management, test, and required installation support are provided.
- v. The Contractor shall conduct site checks and develop/update Installation Drawing (SID) packages, in conjunction with the installations and removals.
- w. The Contractor shall support site inspections, tests, trainings, and troubleshooting actions associated with training systems.
- x. The Contractor shall provide support to test, evaluations, verification and validation events, test documentation preparation, test planning, requirements traceability analyses, data management and analysis plan preparation, test scenario development, support to unit level and multi-element test, test execution support, test data recording and test data analysis.
- y. The Contractor shall conduct engineering studies and analyses of actual and proposed Shore Site Installations to assess physical and functional capability as well as supportability.
- z. The Contractor shall analyze operational and maintenance performance data to determine design defects detracting from the system's capability to perform its intended operational requirements and to meet its maintainability and reliability requirements.
- aa. The Contractor shall support T&E to include test planning, test scenario development, data management and analysis plan preparation, test execution, test data recording, and developmental and operational test & evaluation functions.

- bb. The Contractor shall support equipment set-up, run interface cables, and build cable harnesses.
- cc. The Contractor shall support for Electronic Magnetic Interference (EMI) test and qualification events. This support shall include EMI test planning, EMI analyses, EMI-related technical correspondence, and EMI test event data analysis.
- dd. The Contractor shall conduct engineering studies and analyses of proposed new equipment to assess physical and functional capability as well as supportability.
- ee. The Contractor shall analyze operational and maintenance performance data to determine design defects detracting from the system's capability to perform its intended operational requirements and to meet its maintainability and reliability requirements.
- ff. The Contractor shall provide inputs to engineering changes and alterations to provide required improvements and to correct service deficiencies. The Contractor shall provide support for coordination of acquisition of system components necessary for integration into shipboard training systems.
- gg. The Contractor shall support formal system engineering technical reviews associated with requirements analysis, Architectural Design, Implementation, Integration, Verification, Validation and Transition technical processes.
- hh. The Contractor shall support assessments during new builds, including recommendations for the use of Commercial Off-the-Shelf and Non-developmental items where appropriate.
- ii. The Contractor shall support shock test planning, shock analyses, shock technical correspondence and shock event data analysis.
- jj. The Contractor shall support safety analyses/reviews of proposed engineering changes and new operating and maintenance procedures and changes, ensuring that the procedures, warnings, and cautions are adequate and inherent safety is not degraded. The Contractor shall analyze safety deficiency reports, results of failure analyses and mishap investigations, and recommend corrective actions.
- kk. The Contractor shall review, prepare, and provide inputs for ECP's for configuration impact and provide recommendations to the ITS/ISEA team.
- ll. The Contractor shall provide engineering drawing support: red-lining Production Level Engineering Drawings and associated documents, ECP drawings changes, Technical Manual illustration updates, and field change illustration updates.
- mm. The Contractor shall review and provide recommendations regarding specific support equipment that are required to support the operation and maintenance of systems such as Silver Disks and General-Purpose Electronic Test Equipment.

APPLICABLE CDRL(s):

C.3.5 Cybersecurity

- a. The Contractor shall provide cybersecurity support services to the Government in support of shore-based training objectives.
- b. The Contractor shall provide system engineering to ensure Cyber compliance and the posture with respect to availability, integrity, authentication, confidentiality, and non-repudiation of critical system information.
- c. The Contractor shall generate and maintain requirements to ensure IA controls are defined and validated.
- d. The Contractor shall perform Cyber management functions in compliance with all applicable DOD, DON, SECNAV and OPNAV directives and instructions.
- e. The Contractor shall comply with applicable Cyber Security, certification and accreditation guidance as directed by DOD, DON, SECNAV and OPNAV National Institute of Standards and Technology (NIST), National Initiative for Cybersecurity Education (NICE), NICE Cybersecurity Workforce Framework (NCWF).
- f. The Contractor shall follow the DoN Risk Management Framework mapping of security controls overlays based on the applicable directives, policy, and instructions.
- g. The Contractor shall ensure Cyber attributes are achieved through the cost-effective, risk-balanced application of controls in a manner that promotes confidentiality, integrity, availability, non-repudiation, and authentication of Test to ensure CE rigor has been conducted in the development and operation Life-Cycle of the system.
- h. The Contractor shall support the accreditation process.
- i. The Contractor shall identify, report, and resolve security violations and ensure that all information systems are functional and secure; recommend cybersecurity/security solutions to support requirements.

APPLICABLE CDRL(s):

C.3.6 Test and Evaluation (T&E)

- a. The Contractor shall provide T&E support services to the Government in support of shore-based training objectives.
- b. The Contractor shall support T&E efforts in all phases of system test including Unit, Development, Integration, Acceptance, and Certification.
- c. The Contractor shall develop, maintain, and perform test procedures such as, Functional Operability Tests (FOTs), Warfare Area Tests (WT), Stress Tests, or Endurance tests.

- d. The Contractor shall perform test bed validation to include configuring and documenting test bed configuration and executing test.
- e. The Contractor shall submit daily test reports detailing test activities conducted during scheduled test events.
- f. The Contractor shall submit final test reports which summarize the collection of Daily Test Reports during a given test event.
- g. The Contractor shall generate and submit software deficiency reports (Failure Analysis and Corrective Action Report (FRACAS)) of observed problems during testing.
- h. The Contractor shall support comprehensive test and evaluation associated with test support, operational verification of installations, verification and validation processes including developmental test and evaluation and operational test and evaluation of new/upgraded systems and components.
- i. The Contractor shall support comprehensive test and evaluation associated with the development, maintenance and performance of test procedures.
- j. The Contractor shall develop test reports at the conclusion of test events, generating, and submitting software deficiency reports of observed problems and/or operational verification of installations.
- k. The Contractor shall develop and maintain shipboard operability validation test procedures in support of shipboard installations.
- l. The Contractor shall develop, execute, and maintain unit tests using test tools such as CppUnit, JUnit, Mockito and Google Tests. Use automated test execution tools whenever feasible.
- m. The Contractor shall schedule, coordinate, and support developer and integration test events at local and remote sites.
- n. The Contractor shall develop and maintain developer integration test plans and schedules.
- o. The Contractor shall develop, maintain, and execute developer integration test procedures to validate software functionality, ensuring all requirements are met and all capabilities have been tested.
- p. The Contractor shall conduct software integration planning and testing of computer programs at shore site locations.
- q. The Contractor shall perform system stress and regression testing of software baselines to ensure no degradation of software capabilities.

- r. The Contractor shall validate test procedures and test beds prior to developer integration tests.
- s. The Contractor shall document test results, test coverage, and overall test status.

APPLICABLE CDRL(s):

C.3.7 Life-Cycle Maintenance

- a. The Contractor shall provide software Life-Cycle maintenance services to the Government in support of shore-based training objectives.
- b. The Contractor shall respond to users' requests and inquiries regarding software troubleshooting, usage and debugging.
- c. The Contractor shall provide solutions for technical and project related issues.
- d. The Contractor shall provide responses and recommendations to address possible Cybersecurity and Safety findings.
- e. The Contractor shall generate Software Trouble Reports for observed software problems and defects.
- f. The Contractor shall support software corrections and updates to existing baseline deliveries.
- g. The Contractor shall identify and correct identified problems in accordance with established patch build processes.
- h. The Contractor shall utilize software vulnerability scanners to identify software security risks. The Contractor shall implement configuration changes, software corrections and applying patches to resolve cybersecurity findings and mitigate security risks.
- i. The Contractor shall provide software metrics required in support of software development, Software Life-Cycle maintenance and process improvement including software lines of code, running time complexity, and defect counts.
- j. The Contractor shall identify, recommend, and document lessons learned after each release and make recommendations as appropriate to increase the quality of future deliverables and improve software reliability.
- k. The Contractor shall perform prototypical investigation and development of technical solutions for obsolescence and technical issues as requested.
- l. The Contractor shall research, design, install, test, troubleshoot, and repair integrated training systems' software including shipboard systems and land-based sites on a worldwide basis.

- m. The Contractor shall develop and maintain Software User Manuals (SUM), Software Version Description (SVD), Software Design Notebook (SDN), and Software/Subsystem Design Description (SSDD) and other technical description documentation as required.
- n. The Contractor shall provide support for fleet synthetic training – unit preparations including maintenance training and operator training.
- o. The Contractor shall support the development of system introductory/familiarization and operator/maintainer courses and course materials.
- p. The Contractor shall update and maintain training system databases.
- q. The Contractor shall develop, maintain, and deliver training curriculum and aids, Fleet Introduction Training and providing inputs into Naval Training System Plans (NTSP) and Training Planning Process Methodologies.
- r. The Contractor shall provide support for NTSP development and updates.
- s. The Contractor shall generate, update, and coordinate training system course curriculum throughout the approval process, to ensure schoolhouse course curriculum will pass audits.
- t. The Contractor shall ensure Knowledge Skills and Abilities are identified and covered in course curriculum.
- u. The Contractor shall maintain Training Systems Alteration Record.
- v. The Contractor shall provide training for identified trainees.
- w. The Contractor shall generate, maintain, and deliver Technical Manuals, Technical Manual Contract Requirements, ILS Data Packages, ILS Certifications and checklists, User's Logistics Support Summaries, Maintenance Plans, Maintenance Task Analysis, System Training Plans, Combat Systems Operational Sequencing System, and the Integrated Logistics Support Plans, or other product support documents in accordance with the current build schedule.
- x. The Contractor shall analyze, evaluate, and prepare program briefs and reports.
- y. The Contractor shall respond to shore site requests for technical assistance and identifying assets from various Inventory accounts for installations in the correct configuration of materials to meet the projects install schedule.
- z. The Contractor shall analyze and develop data for sparing models for supportability, maintainability, and reliability.
- aa. The Contractor shall develop and maintain Installation Checkout (INCO) spares lists and maintain records for replenishment of INCO requirements.

- bb. The Contractor shall analyze, maintain, and provide level of repair analyses, failure analysis and corrective action report, reliability centered maintenance analysis, failure mode and effects analysis, failure mode analysis, effects and criticality analysis, or spectrum supportability risk assessments to meet system/subsystem introductions in accordance with the Government provided build plan and schedules.
- cc. The Contractor shall generate and review Technical Data Packages (TDPs), top-down break down lists, letter of authorizations, adaptation files, consolidate overhaul planning files and associated software trouble reports, and nomenclature assignments.
- dd. The Contractor shall resolve discrepancy reports.

APPLICABLE CDRL(s):

C.3.8 Configuration Management

- a. The Contractor shall provide configuration management services to the Government in support of shore-based training objectives.
- b. The Contractor shall develop and manage software configuration management plans, processes, and procedures.
- c. The Contractor shall maintain a software development and configuration environment using tool such as Dimensions, Subversion, Atlassian Jira, Atlassian Crucible, or Fisheye.
- d. The Contractor shall identify and recommend artifacts to be placed under configuration control.
- e. The Contractor shall perform configuration and change management using tool such as Subversion, Jira, Crucible, DOORS, or Microsoft SharePoint of all software development artifacts.
- f. The Contractor shall perform configuration and change management of unit tests, test execution scripts and associated files.
- g. The Contractor shall perform configuration and change management of documentation related to the requirements analysis, design, implementation, and test of software applications including software design documents, version description documents, test plans, test procedures and processes.
- h. The Contractor shall perform configuration audits as required by the established CM plans and procedures.
- i. The Contractor shall track, record, and report status of configuration items and baselines.
- j. The Contractor shall review IRPs, ECPs and other proposed change requests and provide recommendations for final Government evaluation and approval by the appropriate

Configuration Control Board in accordance with the established CM plans and procedures.

- k. The Contractor shall maintain data within DOD/DON-mandated CM/ILS applications.
- l. The Contractor shall prepare and provide recommendations to ILS and Configuration Maintenance plans.
- m. The Contractor shall maintain system baseline documentation, specifications, technical data package, and other data forming the configuration managed product baseline.
- n. The Contractor shall review and update maintenance index pages or maintenance requirement cards, concepts, procedures, and plans.
- o. The Contractor shall provide comment and recommendation for remediation in response to technical feedback reports received from shore sites.
- p. The Contractor shall compile initial configurations, allowance, and technical provisioning/re-provisioning coding data to the Interactive Computer Aided Provisioning (ICAP), Allowance Parts List (APL) and Repairable Item Coding (XRIC) Systems.
- q. The Contractor shall coordinate provisioning meetings to review and finalize all requests.
- r. The Contractor shall analyze and develop data for sparing models for supportability, maintainability, and reliability.
- s. The Contractor shall compile and maintain program support data, item unique identification tagging and database, transaction items reporting, design interface and system related TDPs, research obsolescence issues and provide recommended solutions, which include parts forecasting, and national item identification number requests.
- t. The Contractor shall review and update CM changes, prepare reports, plans, metrics, and schedules.
- u. The Contractor shall coordinate and conduct shore site configuration audits and analyze equipment configuration files for erroneous data and submit corrections.

The Contractor shall develop and maintain system and equipment interconnect and functional diagrams, network architecture and facility drawings and diagrams utilizing Computer-Aided Design (CAD) software applications.

- v. The Contractor shall incorporate all equipment deletions, additions or cabling modifications performed on any of the systems or facilities into the diagrams and drawings.

- w. The Contractor shall derive data for specific changes requiring drawing modification from a Site Change Implementation Notice (SCIN), training site activation plans and periodic physical audits.
- x. The Contractor shall perform and capture detailed system activation and installation planning via SCIN.
- y. The Contractor shall ensure SCINs contain such details as a work breakdown structure, facility and configuration diagrams, equipment delivery schedules and storage locations, equipment removal and installation sequences, and cable installation sequences.
- z. The Contractor shall provide activation support including cable staging, configuration monitoring and verification during installation.
- aa. The Contractor shall conduct post-activation analysis to identify lessons learned and process improvements for future activations.

APPLICABLE CDRL(s):

C.3.9 Lab Support

- a. The Contractor shall provide Technical Representative laboratory support services to the Government in support of shore-based training objectives.
- b. The Contractor shall provide set-up and maintenance of laboratory equipment and workstations.
- c. The Contractor shall provide management of hardware and software licenses associated technical support information.
- d. The Contractor shall provide identification of materials needed for performing lab operations including hardware, software, tools, and consumables.
- e. The Contractor shall generate and maintain laboratory and network configuration drawings.
- f. The Contractor shall provide support for lab demonstrations and other meetings.
- g. The Contractor shall maintain Lab inventory (OM&S) in Enterprise Resource Planning (ERP).
- h. The Contractor shall support ITS Division test, evaluation, verification, and validation events. This support includes, test documentation preparation, test planning, requirements traceability analyses, data management and analysis plan preparation, test scenario development, support to unit level and multi-element test, test execution support, test data recording, test data analysis, other support to developmental and operational test events.

- i. The Contractor shall prepare Cybersecurity Accreditation packages and submit into Enterprise Mission Assurance Support Service (eMASS).
- j. The Contractor shall initiate and track Information Technology Procurement Requests (ITPRs) relating to ITS lab equipment from start to completion.
- k. The Contractor shall perform lab access control and maintain access list procedures and related records.
- l. The Contractor shall perform data backups, disaster recovery operation and maintain disaster recovery plans.

APPLICABLE CDRL(s):

C.3.10 Shore Based Training System Hardware Engineering

- a. The Contractor shall provide training system hardware engineering services to the Government in support of shore-based training objectives. The support duties and responsibilities include tasks such as the following:
- b. The Contractor shall provide hardware and prototype recommendations with an emphasis on using modular design, standards-based interfaces, and widely supported consensus-based standards.
- c. The Contractor shall perform analysis and provide recommendations to the Government subject matter experts to identify and schedule activities for system upgrades/modifications and new system installations.
- d. The Contractor shall provide logistics support for shore site materiel requirements.
- e. The Contractor shall develop and submit appropriate hardware and software trouble reports.

APPLICABLE CDRL(s):

C.3.11 C6ISR: Command, Control, Communications, Computers, Cyber-Defense and Combat Systems and Intelligence, Surveillance, and Reconnaissance

- a. The Contractor shall support C6ISR engineering services to the Government in support of shore-based training objectives.
- b. The Contractor shall review Shore Based Training Equipment documentation for impacts for C6ISR and provide recommendations for improvements.
- c. The Contractor shall review system level baseline build plans, identify/track, and plan for support needed for plan baselines, and assist in project scope management and integration efforts for C6ISR requirements.

- d. The Contractor shall support and maintain the C6ISR Integration and Interoperability facilities infrastructure and supporting work areas with reconfiguration as required.
- e. The Contractor shall conduct engineering studies and analyses of communication and network security equipment to assess physical and functional capability as well as supportability.
- f. The Contractor shall support the implementation of boundary defense architecture including firewalls, intrusion detection/prevention systems, encryptions, and future technologies as they evolve.
- g. The Contractor shall support C6ISR T&E, to include test planning, test scenario development, data management and analysis plan preparation, test execution, test data recording, and support of developmental and operational test & evaluation functions.
- h. The Contractor shall recommend corrections or alterations to provide required improvements to correct service deficiencies or improvements for translation into detailed engineering changes proposals; assist in the conduct of design reviews during detail design, construction, and implementation.
- i. The Contractor shall provide support for C6ISR fleet synthetic training network integration efforts.
- j. The Contractor shall provide engineering support and assessments of C6ISR concerns during new builds.
- k. The Contractor shall provide support to developmental test, evaluation, verification and validation of communication and network security equipment events.
- l. The Contractor shall verify correct Firmware and Key mat and proper switch configuration for Systems encryptions.

APPLICABLE CDRL(s):

C.3.12 Material Analysis

- a. The Contractor shall provide material support including procurement, preparation, testing, assembly, and packaging of items required to prototype, properly field new and/or development software, hardware applications, systems/subsystems, and components.
- b. The Contractor shall procure equipment, parts, hardware, and/or software as necessary for installations, maintenance, prototype, Engineering Duty Model (EDM), and or development of systems and components.
- c. The Contractor shall prepare cost estimates for all procurements, shipping requests, and development of the requirements for obsolescence management.

- d. The Contractor shall develop, review and update Supplemental Provisioning Technical Documents or other associated documents as required.
- e. The Contractor shall be responsible for configuration management for new and/or upgraded equipment by maintaining system databases.
- f. The Contractor shall ship and receive equipment and materials that are being transferred to and from ships, shore sites, OEM facilities, and Government locations.

APPLICABLE CDRL(s):

C.3.13 Training Site Installation

- a. The Contractor shall conduct engineering studies and analysis of actual and proposed shore site installations and will develop Training Support Installation Packages.
- b. The Contractor shall assess the physical, functional and support capability of the location and will perform facility visits to verify dimensions, interfaces, power availability, tolerances and other installation factors
- c. The Contractor shall perform preliminary installation checkouts such as cable continuity checks, visual inspections, connector pin verification and operational verification.
- d. The Contractor shall participate in installation meetings and will support the development of hardware and software installation plans, parts lists, installation instructions, checkout plans, test procedures and drawings as required to complete shore site installations.
- e. The Contractor shall review and comment on installation standards and practices.
- f. The Contractor shall examine installation design and physical layout for reliability, ease of maintenance, and suitability to perform equipment and system operational performance requirements and develop corrective modifications and alterations.
- g. The Contractor shall support the coordination of shore site installations and engineering change scheduling for training systems. This includes leading installation efforts, obsolete equipment rip-outs, overhauls, hardware and software upgrades, fabrication of necessary components and assemblies and technical assistance on assemblies, units, equipment, systems, engineering changes and alterations.
- h. The Contractor shall perform post installation checkouts, including interoperability and operational testing.
- i. The Contractor shall resolve any installation and safety related issues and concerns and will report all findings to the Government.
- j. The Contractor shall provide support to test, evaluation, verification and validation events; test documentation preparation, test planning, requirements traceability analysis, data management and analysis plan preparation and test scenario development.

- k. The Contractor shall perform space modifications as required including the installation and sustainment of infrastructure. This support encompasses hardware, foundations, cabling (copper and fiber optic), connector installation, false decking and overhead, deck covering, IT Systems, electrical, heating, ventilation, air conditioning and electronic cooling water piping.
- m. The Contractor shall generate and execute procedures in support of facility activations, upgrades and repairs: provide design, assembly and fabrication, provide the layout and fabrication of equipment, perform the installation and initial verification of equipment and cables installed in the facility.
- n. The Contractor shall determine, analyze, recommend and implement environmental solutions to ensure that the services being provisioned to the lab support the lab space final configuration.
- o. The Contractor shall support all stages associated with lab space development as outlined in the Training Installation and Transfer Agreement (TITA) to ensure that upon completion the lab space operates IAW end user requirements.
- p. The Contractor shall perform facilities modifications in order to meet the final environmental requirements needed to allow lab operation in its final configured state.

APPLICABLE CDRL(s):

C.3.14 Training Site Accreditation

- a. The Contractor shall support in the development and execution of accreditation plans for complex test sites and simulation systems for both hardware-in-the-loop and virtual testing as a system under test within a certified accreditation boundary. This may include providing Information Assurance Officer services and applying tactical computer program software, installations, patches and updates.
- b. The Contractor shall provide for multiple platform integration of various geographically disbursed test beds to fully integrate elements in distributed, multisite test events.
- c. The Contractor shall provide ongoing technical expertise to perform the integration, configuration, operation and accreditation of the laboratory test platforms over classified and unclassified networks and communication systems.
- d. The Contractor shall be responsible for establishing, monitoring, troubleshooting and supporting all unclassified, classified and tactical support networks in support of testing for traffic levels, identification of problem areas, and immediate response to network alarm conditions within the training site.

- e. The Contractor shall ensure alarm conditions include excessive collisions, backbone failure, remote link failure, network equipment failure or power failure.

APPLICABLE CDRL(s):

C.3.15 Training Site Operations

- a. The Contractor shall provide onsite operational support to facility staff and users and execute hardware and software maintenance.
- b. The Contractor shall provide direct user interface support: develop and implement test plans, metrics collection, and load required media, tactical and simulation systems required to complete test objectives.
- c. The Contractor shall initialize and secure the training site in accordance with the Government Furnished Instruction startup/shutdown procedures.
- d. The Contractor shall prepare the trainer into a Ready-For-Training state in support of scheduled ad-hoc-military training sessions or other operational use by the Government.
- e. The Contractor shall determine the functional status of equipment and conduct equipment maintenance as required to include; troubleshooting hardware and software faults, ordering and installing replacement parts, and conducting preventative maintenance in accordance with the Government established Preventative Maintenance Schedule.
- f. The Contractor shall maintain onsite spares inventory to perform same day repairs of identified high failure items and have the capability to deliver failed items within 48-hours.
- g. The Contractor shall ensure all corrective and preventative maintenance actions are documented and reported to the Government site manager(s).
- h. The Contractor shall provide system safety engineering support to include safety program planning, system safety engineering analyses, definition of safety requirements, safety verification and validations of designs, implementations, and other mitigations.
- i. The Contractor shall identify safety risks and document in Government established Hazard Assessment reports.
- j. The Contractor shall provide general facility experience to coordinate minor construction, facility refurbishment, and maintenance activities.
- k. The Contractor shall coordinate minor maintenance actions including key-and-lock control, abatement of safety infractions, and logistical support for general facilities maintenance.
- l. The Contractor shall maintain oversight of power management equipment including, uninterrupted power supplies, back-up generator systems, transfer switches, HVAC

systems, cooling water systems and other industrial systems and functions not executed by Public Works or Naval Facilities Command.

- m. The Contractor shall be expected to provide after-hours support for facility issues.
- n. The Contractor shall provide Personal Protective Equipment (PPE) required for all duties.
- o. The Contractor shall not perform any functions executed by Public Works or Naval Facilities Command.
- p. The Contractor shall develop and maintain accurate engineering drawings, facility Standard Operating Procedures (SOPs), space drawings and layouts to include critical infrastructure.
- q. The Contractor shall administer a lock-out/tag-out program that supports safe execution of maintenance across all training sites and facilities.
- r. The Contractor shall coordinate recurring maintenance and inspections on project assets such as UPS systems, processed water systems, 400Hz converters and other equipment.
- s. The Contractor shall coordinate all support services required to support training site activations and building activities.
- t. The Contractor shall install, setup, maintain, troubleshoot, and train users and breakdown after the use of conference room support devices such as video teleconference (VTC), webcasts, computers and overhead projector devices.

APPLICABLE CDRL(s):

C.4 TASK ORDER MANAGEMENT

The Contractor shall provide sufficient supervision to ensure all personnel work instructions and schedules originate with the Contractor rather than with the Government technical office. This supervision shall ensure that the Contractor's personnel are not assigned work instructions and schedules outside this SOW. Any perceived conflicts with the SOW shall be directed to the Contracting Officer and Contracting Officer's Representative (COR).

C.4.1 Progress Report

The Contractor shall submit a Contracting Officer's Management Report. **(CDRL A00X)** The Contractor shall maintain and provide a list of all personnel assigned to the Task Order. The Contractor shall provide copies of invoices received from Sub-contractors and consultants as well as evidence of its review to ensure that all invoiced costs represent services that have been performed and deliverable items. In accordance with G-232-H005 and FAR 52.216-7, the Contractor shall provide the COR supporting documentation, including receipts, for all travel and material purchases with each invoice submission.

C.4.2 In Progress Reviews (IPRs)

The Contractor shall participate in both formal and informal In Progress Reviews (IPRs) of work being performed. IPRs may be conducted at the overall Task Order level or for specific projects/work areas. Reviews will be scheduled by the Government. An initial formal IPR is planned within 90 days of the Task Order award date and will follow the agenda agreed to by the COR. Subsequent formal IPRs shall be held approximately every six (6) months thereafter unless waived by the Government. The IPRs shall be conducted at a location mutually agreed upon by the Contractor and the Government. At the time the review is scheduled, the Government will communicate the specific purpose of the review and advise the Contractor as to the desired content of the presentation. **(CDRL A00X)**

C.4.3 Task Prioritization Meetings

The Contractor shall participate in periodic Task Prioritization Meetings with the COR and Subject Matter Experts (SMEs). The purpose of these meetings is to convey the Government's technical program schedules and priorities and to identify corresponding task priorities for this Task Order. The Contractor shall document meeting minutes of the meetings. **(CDRL A00X)**

C.4.4 Telework

In accordance with H-216-H002 Level of Effort -- Alternate I (NAVSEA) teleworking at an alternative worksite is permitted. Telework is also dependent upon the type of work being performed. Restrictions against teleworking apply to those labor categories with security concerns and restrictions related to classification of the data being administered, or those labor categories with the type of work that is required to be performed full-time at a Government facility location. The following labor categories already specified in Section C.7.1 of this SOW are not authorized for telework and are required to perform their duties full-time at the designated Government facility location. The following labor categories require hands-on access to lab equipment and systems.

Labor Category
Systems Technician 1 HH 218-221
Systems Technician 2 HH 218-221
Systems Technician 3 CEP 163 (NOB)
Systems Technician 2 CEP 163 (NOB)
System Administrator 2 HH 218-221
Systems Administrator 2 CEP 163 (NOB)

Systems Technician 1 San Diego
Systems Technician 2 San Diego
Systems Technician 3 San Diego
Systems Technician 1 Rota, Spain
Systems Technician 2 Rota, Spain
Systems Technician 2 Yokosuka, Japan
Systems Technician 1 Yokosuka, Japan
Systems Technician 2 Everett
Systems Technician 1 Everett

C.4.5 Technical Instructions (TIs)

- a. The Government will authorize all work to be performed under this Task Order through the issuance of formal Technical Instructions (TIs) signed by the Contracting Officer or their designated representative.
- b. Each TI will provide specific technical requirements, objectives, deliverables, and periods of performance for a discrete task. All TIs will fall within the overall scope and disciplines described in Section C.3 of this SOW.
- c. The Contractor shall not proceed with any work or incur any costs until a TI for that work has been formally issued. Any work performed by the Contractor outside the scope of an approved TI is done at the Contractor's own risk and may not be considered for payment.

C.5 SKILLS AND TRAINING

C.5.1 The Contractor shall provide capable personnel with qualifications, experience levels, security clearances, and necessary licenses, certifications, and training required by Federal, State, and Local laws and regulations. Information assurance functions require certifications specified in Defense Federal Acquisition Regulation Supplement (DFARS) 252.239-7001 Information Assurance Contractor Training and Certification. The Contractor shall ensure the performing employee is trained/certified in IA commensurate with their level/category of access.

C.5.2 Per the requirements set forth in DOD 8570.01-M Information Assurance Workforce Improvement Program, or successor, personnel performing Information Assurance (IA) functions outlined in the manual will be tracked for compliance, designated in writing from the Local Command Authority as Cyberspace Workforce (CWF), shall already be in possession of

an approved commercial certification for their environment at time of performance start (i.e. end of the 30-day transition period), and shall satisfy both preparatory and sustaining DOD IA training and certification requirements as outlined in Chapters 2-5 of DOD 8570.01-M. Additionally, personnel with Privileged Access must sign acknowledging duties on a "Privileged Access Agreement", be able to maintain the appropriate investigation type, sign a System Authorization Access Request-Navy (SAAR-N) with "privileged" indicated, complete applicable Operating System/Computer Environment (OS/CE) training to Navy's standards, and maintain a valid commercial certification that meets outlined designations. Failure to maintain all requirements at the appropriate level may result in the removal of elevated access and removal from tasks being performed. Non-privileged Information Assurance Workforce (IAWF) members may be removed from tasks being performed. In both situations, access will be returned once the failure(s) are corrected.

C.5.3 Training necessary to ensure that personnel performing under this Task Order maintain the knowledge and skills to successfully perform the required functions is the responsibility of the Contractor. Training necessary to maintain professional certification is the responsibility of the Contractor. The Contractor shall provide training opportunities to maintain Contractor capabilities and/skills to coincide with changing technology environments. All costs for DOD 8570.01-M certification shall be borne by the Contractor for new employees on this Task Order.

- a. Cyber IT/CSWF members are required to complete the requisite quantity of continuing education training, also referred to as continuous learning, currently forty (40) hours every year in addition to the requirement to completing Annual Cyber Awareness training as directed by the DON. Cyber IT/CSWF program qualification is a requirement of all positions that are involved in the cybersecurity of computers, systems, and or networks.

C.5.4 Training on non-commercial and department-specific tools and procedures will be provided by the Government.

C.5.5 Notwithstanding the above, it is understood that the Government may require services requiring knowledge and skills of a newly emergent technology. The Contracting Officer may, on a case-by-case basis, authorize the expenditure of Government funds for the training of Contractor personnel. Advance approval is required. The Contractor is encouraged to suggest a cost-sharing arrangement that addresses registration, tuition, travel, and labor costs.

C.5.6 The Contractor shall complete all mandatory training as prescribed by the Government. These training requirements are accessed through Government web-based training (WBT), classroom training, or combination of both. All Contractor personnel shall be qualified and maintain certifications as prescribed throughout this SOW and in accordance with DODD 8750.01 at time of contract award, on the first day of performance, and throughout the duration of the contract.

- a. Repair and maintenance employees working aboard vessels, dry docks, and piers shall have a valid 10-hour OSHA Maritime Shipyard Employment Course #7615 or NAVSEA-approved equivalent completion card within 60 days of employment.

- b. Private Shipyards. Private shipyards may require other documentation, liability insurances, and additional training in advance of proceeding to the yard.
- c. There may be costs associated with any training required for Contractors to access private shipyards. Prior to incurrence of costs, COR approval is required for these costs to be chargeable to the contract.

C.5.7 Mandatory Training Completion Log - The Contractor shall submit a CWF report (CDRL AXXX). The CWF report shall address required CWF training including annual continuous learning training requirements. The Contractor shall also submit Mandatory Training Completion Log (CDRL AXXX). The Mandatory Training Completion Log shall address all mandatory training completed as directed by the COR.

- a. The report shall include employee name, TI#, the applicable Information Assurance category/specialty and level (see SOW or TI), required certifications and fulfillment status. New hire information for tasking requiring cyber functions shall be submitted to the COR at least seven (7) days prior to employee beginning performance of any cyber functions. New hire information shall include name, TI#, list of applicable cyber functions category, level, required certifications and fulfillment status to include a copy of the baseline certification documentation.

C.6 OTHER DIRECT COSTS

During the performance of this Task Order the Contractor will have to travel and procure material and consumables to perform the tasking of the SOW.

C.6.1 Travel

The Contractor shall be required to travel in performance of this Task Order. The numbers of trips and types of personnel traveling shall be limited to the minimum required to accomplish work requirements and shall be requested of and authorized in advance by the COR by email. All requests shall show the appropriate Task Order number, the number of people traveling, the number of days for the trip, the reason for the travel, an estimated cost (with G&A separate), and any high cost or unusual costs expected. The Contractor is not authorized to perform any travel that is not in conjunction with this effort. All travel shall be conducted in accordance with Federal Acquisition Regulation (FAR) 31.205-46 Travel Costs. Locations will be determined by the Government and are subject to change.

Representative travel locations may include:

- Virginia Beach, VA
- Wallops Island, VA
- Norfolk, VA
- Dahlgren, VA
- Washington, DC
- Patuxent River, MD
- Pearl Harbor, HI
- San Diego, CA
- Mayport, FL
- Orlando, FL
- Groton, CT
- Great Lakes, IL

- Bremerton, WA
- Spain
- South Korea
- Japan
- Australia
- Philippines
- Guam
- Italy
- Bahrain
- Alaska
- Taiwan
- New Zealand
- Germany
- Canada
- United Kingdom
- Poland
- Romania
- Israel

The Contractor shall submit trip reports following any travel under this effort. (CDRL AXXX)

C.6.2 Direct Charge of Materials

- a. During the performance of this Task Order it may be necessary for the Contractor to procure materials to respond to the mission requirements listed in the SOW. This Task Order is issued under a services contract and the procurement of materials of any kind that are not directly related to and necessary for performance may be determined to be unallowable costs pursuant to FAR Part 31. Any material provided by the Contractor is subject to the requirements of the FAR, DFARS, and the applicable Department of Navy regulations and instructions. Charges related to materials costs may include general and administrative (G&A) expenses but shall not include fee or profit.
- b. The materials listed in paragraph (e) below shall be the only ones approved for purchase under this Task Order. The numbers of, and types of, material purchases shall be limited to the minimum required to accomplish work requirements and shall be coordinated with the COR. Splitting purchase requirements to defeat purchasing thresholds shall not be approved. If the Contractor (or Sub-contractor) possesses a Defense Contract Management Agency (DCMA) approved purchasing system, individual item purchases greater than \$3,000 but less than \$10,000 shall be approved by the COR; individual item purchases equal to or over \$10,000 shall not be executed until the COR reviews the requested purchase and approval is obtained from the Contracting Officer. If the Contractor does not possess an approved purchasing system, individual item purchases less than \$3,000 shall be approved by the COR. For purchase of materials equal to or over \$3,000 shall not be executed until the COR reviews the requested purchase and approval is obtained from the Contracting Officer. No purchases of any amount shall be executed by a Sub-contractor if they do not have a Defense Contract Audit Agency (DCAA) approved accounting system.
- c. Information Technology (IT) Resources shall not be purchased unless DOD and Navy purchasing procedures (including Information Technology Procurement Request (ITPR) approval) have been satisfied and approvals obtained. IT Resources include personal computers (PCs), laptops, printers, software, servers, hubs, routers, phones, fax machines, and any related maintenance, telecommunications, training, or other support services. All IT Resource Other Direct Cost (ODC) purchases require COR approval regardless of the dollar value associated with the purchase.

- d. The costs of general purpose business expenses required for the conduct of normal business operations will not be considered allowable direct costs in the performance of the Task Order - unless supported by DCMA as a procedure of the Contractor's accounting procedures. General purpose business expenses include the costs for items such as telephones (including cell phones) and telephone charges, copy machines, word processing equipment, personal computers, and other office equipment and supplies.
- e. The materials and equipment contained in the following list are authorized for purchase once the Contractor has complied with approval requirements stated in paragraph (b) above and paragraph (f) below:

- 1) IT Hardware and software products (computers, servers, switches, hard drives, firewalls, cables, applications, development environments and other similar types of items), Engineering Development Model (EDM) Equipment, Components, Subcomponents and piece parts; Software Licenses (e.g. Unity, Microsoft, Tableau, Auto Desk Inventor, Visual Studios, Marmoset, and similar types of software, applications, development environments and/or licenses).

Materials not fitting the descriptions listed in this section may not be purchased unless mutually agreed to by both the Government and Contractor and subsequently added to the contract by Task Order modification.

- f. Procedure for obtaining COR and/or Contracting Officer approval, the Contractor shall submit a request for purchase of materials to the COR through email. The COR will review the request, if it is in accordance with paragraph (b) above and requires Contracting Officer approval, the COR will submit the request via the Contract Specialist to the Contracting Officer for review and approval. Minimum requirements for a written request for purchase are as follows:
 - 1) Description of the material to be purchased;
 - 2) Quantity;
 - 3) Unit and total cost;
 - 4) Delivery/freight charges;
 - 5) Any associated service charges such as assembly, configuration, packing, etc.;
 - 6) An explanation of the need for the material;
 - 7) Copy of the competitive quotes received from potential suppliers;
 - 8) The basis for the selection of the selected supplier;
 - 9) Explanation of the determination of price reasonableness regarding the selected supplier costs;
 - 10) If the procurement is sole sourced to a particular supplier or manufacturer, include the rationale for limiting the procurement to that supplier or manufacturer;
 - 11) ITPR request date, and if granted, approval date and tracking number (if applicable);

Once the COR and/or Contracting Officer have reviewed the request, the Government will notify the Contractor of the outcome. Issues or details may be discussed with the

Contract Specialist acting on behalf of the Contracting Officer until a final Government determination is made as to whether to approve, modify, or reject the purchase.

- g. The Contractor is required to possess and maintain an adequate property management system. The system must track all materials/ODCs associated with this Task Order. Upon completion of the Task Order Period of Performance, all material associated with this Task Order that were purchased by the Contractor and not depleted or previously delivered to the Government during the performance of this Task Order shall become the property of the Federal Government. The Contractor shall transfer all materials, not depleted or previously delivered to the Government, to the COR by way of a Material Inspection and Receiving Report (DD Form 250) in accordance with DFARS PGI 245.402-71 Delivery of Contractor-Acquired Property.

C.6.3 Counterfeit Parts and Materiel

- a. The following minimum processes shall be implemented and flowed down by the Contractor to all Sub-contractors in order to minimize the risk of use of counterfeit parts.
- b. The Contractor shall establish processes to minimize the risk of procuring and using counterfeit parts and materiel. The Contractor shall document these processes and provide those documented processes to Government representatives upon request. The Contractor shall ensure that all requirements are flowed down to Sub-contractors and suppliers at all tiers.
- c. At a minimum, these processes shall ensure that:
 - 1) Materiel that is either in production or currently available shall be purchased from an original manufacturer, aftermarket manufacturer, or other authorized supplier.
 - 2) Materiel that is neither in production nor currently available may be purchased from suppliers, including unauthorized suppliers, that meet appropriate counterfeit avoidance criteria documented in industry anti-counterfeit standards. The Contractor shall notify the KO and COR when critical or high-risk materiel is procured from an unauthorized source. The Contractor shall provide this notification before materiel is installed so mitigation actions can be taken.
 - 3) The Contractor shall generate a Counterfeit Prevention Plan in accordance with DI-MISC-81832 Counterfeit Prevention Plan, and the requirements of this Data Item Description (DID) and SAE Aerospace Standard 5553 (SAE AS5553) Counterfeit Electronic Parts; Avoidance, Detection, Mitigation, and Disposition (CDRL A00X).
 - 4) The Contractor shall maximize the use of authentic, originally designed and/or qualified parts.
 - 5) The Contractor shall assess potential suppliers to minimize the risk of receiving counterfeit parts or materiel.
 - 6) The Contractor shall have purchasing procedures which confirm whether a selected supplier is authorized (as defined in SAE AS5553) for each purchase.
 - 7) The Contractor shall define minimum inspection and test requirements for parts being procured and shall ensure that in-house, third-party, and/or supplier inspection and test procedures and facilities comply with the requirements of this section. These

- minimum inspection and test requirements shall specify appropriate test methods to detect potential counterfeit parts and materiel.
- 8) The Contractor shall require a certificate of conformance (as defined in SAE AS5553) and supply chain traceability for all electronic part purchases.
 - 9) The Contractor shall use government or industry services such as Government-Industry Data Exchange Program (GIDEP) and other commercially available services to identify part or supplier quality or authenticity problems.
 - 10) The Contractor shall maintain critical or high-risk materiel traceability that enables tracking through the supply chain to the original manufacturer if purchased from an authorized source or to the unauthorized source at a minimum if purchased from an unauthorized source.
 - 11) For maintenance and repair efforts, the Contractor shall assess materiel for criticality or high-risk of counterfeiting. The Contractor shall document materiel identified as critical or high-risk. The Contractor shall document mitigating actions to ensure that authentic critical or high-risk materiel is procured and installed.
- d. The Contractor shall immediately notify the COR of the occurrence of a suspect or confirmed counterfeit part or materiel. Contractor shall notify the COR of all actions taken to identify, contain, and impound all product from the lot, within 7 working days of confirmation of the counterfeit status. The Contractor shall initiate and submit an ALERT to GIDEP within 60 days of knowledge of the counterfeit part or materiel.
- e. Suspect counterfeit parts and materiel shall be impounded with all other items from the same lot. The Contractor shall identify and locate all potential users or hardware items with the suspect part or materiel and contain the product which has this suspect part or materiel, pending confirmation of the part or materiel's authenticity. The Original Equipment Manufacturer (OEM) may be involved at this point to verify authenticity. Confirmed counterfeit materiel shall be contained and provided to investigative agencies for ongoing investigation or prosecution. The counterfeit product shall not be scrapped or otherwise disposed of without approval from investigative authorities or the Contracting Officer. Confirmed counterfeit product shall be disposed of in a way which would not allow its resale or reuse.
- f. Counterfeit parts are defined in SAE AS5553. Counterfeit parts may be electronic or mechanical in nature. Counterfeit electronic parts may typically be used parts which have been refurbished and represented as new. Commonly counterfeited electronic parts include parts such as microcontrollers or specially screened devices, or common parts, which have several pin-compatible versions from multiple manufacturers, such as memory devices and operational amplifiers. Counterfeit mechanical parts are typically improperly made, marked, or treated products. Examples are improper anodization or heat treatments (or falsified data), mismarked parts sold as higher grade steel, or used/fake parts such as valves or circuit breakers.

C.7 GOVERNMENT FURNISHED SPACE/GOVERNMENT FURNISHED PROPERTY/GOVERNMENT FURNISHED INFORMATION

C.7.1 Government Furnished Space

The Government requires allocation of personnel to the following Government facility locations and reserves the right to make adjustments based on Government needs at time of tasking execution.

- a. Access to Government Lab Space will be provided. Lab space does not constitute office space. Government Furnished Lab Space will be provided for the following labor categories and number of personnel. No separate office space will be provided.

Labor Category	Number
Systems Technician 1 HH 218-221	1
Systems Technician 2 HH 218-221	1
Systems Technician 3 CEP 163 (NOB)	2
Systems Technician 2 CEP 163 (NOB)	2
System Administrator 2 HH 218-221	1
Systems Administrator 2 CEP 163 (NOB)	1

- b. Other Locations of Performance: Other (non-local) Government installations will be the primary work location for the following positions.

Labor Category	Number
Systems Technician 1 San Diego	1
Systems Technician 2 San Diego	1
Systems Technician 3 San Diego	1
Systems Technician 1 Rota, Spain	1
Systems Technician 2 Rota, Spain	2
Systems Technician 2 Yokosuka, Japan	1
Systems Technician 1 Yokosuka, Japan	1
Systems Technician 2 Everette	2
Systems Technician 1 Everette	1

- c. Government Furnished Property (GFP) (hardware, software, firmware) will be provided to support access to NMCI/other Government computer resources as required during performance of this Task Order.

C.7.2 Government Furnished Property (GFP)

- a. GFP may be issued under this Task Order. The Government will make available the Government Property identified in the attachment J.X, Government Furnished Property on a rent-free basis in performance of the Task Order. The Contractor is responsible for all costs related to making the property available for use, such as payment of all transportation, installation, or rehabilitation costs.

- b. The Contractor shall maintain and report the inventory of GFP (**CDRL A00X**) issued under this Task Order. This requirement is in addition to anything required by regulation, statute, or the assigned Government Property Administrator.

C.7.3 Government Furnished Information (GFI)

- a. The Government will provide access to information and documentation required for Task Order performance (Attachment J.X, Government Furnished Information).
- b. All information and documentation shall be retained at the Government work site.

C.8 SECURITY

C.8.1 Personnel and Facility Clearance Requirements

- c. Facility Clearance Level (FCL) Requirement: The Prime Contractor supporting this Task Order is required to have an FCL of TOP SECRET and safeguarding clearance level of SECRET at the primary facility supporting this Task Order.
- d. Personnel Clearance Level (PCL) Requirement: All personnel directly supporting this Task Order, except Management and Support personnel, are required to have, as a minimum, a current SECRET security clearance in the Defense Industrial Security System (DISS). Interim clearances are acceptable. Personnel performing in the following labor categories shall have a final TOP SECRET security clearance with Sensitive Compartmented Information (SCI) eligibility in DISS: Fully Qualified Navy Validator 2 (1 FTE); Information System Security Manager 2 (1 FTE); Software Engineer 2 (2 FTE).

C.8.2 The Contractor will have access to information and spaces with a SECRET. All deliverables associated with this Task Order are unclassified unless otherwise specified. Access to classified spaces and material and generation of classified material shall be in accordance with the attached DD Form 254 (Attachment J.X).

C.8.3 The Department of Defense Contract Security Classification Specification (DD Form 254) provides the security classification requirements for this effort. The Contractor shall obtain facility and personnel security clearances as required by the National Industrial Security Program prior to starting to work on tasks requiring clearances. Access to classified spaces and material and generation of classified material shall be in accordance with the National Industrial Security Program Operating Manual (NISPOM) 32 Code of Federal Regulation (CFR) Part 117 and the NSWCDD Command Security Manual.

C.8.4 Some work accomplished under this Task Order will be classified up to the level of TOP SECRET. The Contractor shall maintain the capability to store information up to and including SECRET. All data generated under this Task Order shall be classified in accordance with the guidance in the NISPOM 32 CFR Part 117 and the NSWCDD Command Security Manual.